

Yazdan Ahmad Qadri

ASSISTANT PROFESSOR

School of Computer Science and Engineering, Yeungnam University, Republic of Korea

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Research Impact

- Inventor of an Internet of Things-based Adaptive Deep Brain Stimulation system for Parkinson's tremor mitigation. Co-inventor in six patents.
- Published as first author in top-ranked journals in the computer networks category, including a paper with ~ 13000 full-text reads and 600+ citations [IEEE Xplore] in IEEE Communication Surveys and Tutorials.
- Awarded "excellent performance team" in academic research supporting programs from the National Research Foundation, Ministry of Education on Dec 16, 2021*.

* Awarded to PI based on my first author publication.

Research & Teaching Experience

Assistant (Research) Professor

Gyeongsan-si, Republic of Korea

YEUNGNAM UNIVERSITY

Mar. 2024 - Present

- **Goals:** Engage in independent research to publish in high-quality research journals and teach courses.
- **Impact:** 1+ year experience as an assistant (research) professor with a research focus on artificial intelligence, wireless communication, and ICT healthcare convergence.
- **Courses:** Artificial Intelligence Applications (Fall'24), Special Topics on Machine Learning (Spring'25).

Postdoctoral Research Associate

Yongin-si, Republic of Korea

KYUNGHEE UNIVERSITY

Mar. 2023 - Feb. 2024

- **Principal Investigator:** Haejoon Jung Ph.D.
- **Primary research focus:** Enabling real-time wireless communication for critical applications.
- **Impact:** Published two peer-reviewed journal articles one each in IEEE Network and IEEE Internet of Things Journal.

Research Internship

Barcelona, Spain

i2CAT

Jul. 2022 - Aug. 2022

- **Designed a custom NS-3 module for IEEE 802.11be WLANs**
Contributed to developing a custom NS-3 extension for simulating Multi-Link Operation in the IEEE 802.11be WLANs. The project was a collaboration with i2CAT and Universitat Politècnica de Catalunya, Barcelona, Spain.
- Internship supervised by Eduard Garcia-Villegas and Daniel Camps Mur.

Academic Qualifications

PhD in Information and Communication Engineering

Gyeongsan-si, Republic of Korea

YEUNGNAM UNIVERSITY

Mar. 2018 - Feb. 2023

- **Thesis Advisor:** Prof. Sung Won Kim.
- **Doctoral Thesis Title:** *Low-Latency Communication over IEEE 802.11 WLANs for Time-Sensitive Applications.*
Investigated the efficacy of orthogonal frequency-division multiple access (OFDMA) to enable real-time communication for time-sensitive applications.
- Co-inventor of **six patents** during my PhD program.

Teaching Experience

- **Teaching assistant** for (0420) *Engineering Management* focused on optimization techniques.
- Contributor to designing and preparing course content for graduate-level **NS-3** simulator course.
- **Key Skills:** Wireless Communication, IEEE 802.11, 5G, NS-3, Cooja, Reinforcement Learning, C++, and Python.

B. Tech-M. Tech [Integrated Degree] in Electronics and Communication Engineering

L. P. UNIVERSITY

Punjab, India

Aug. 2011 - Jun. 2016

- **Thesis Advisor:** Vineet Kumar
- **M. Tech Thesis Title:** *Early Detection of Neurological and Motor Disorders using Signal Processing Techniques.*
Investigated the role of signal processing techniques and supervised learning to detect epilepsy using electroencephalogram (EEG) as an input. Utilized Mel-frequency cepstral coefficients for feature extraction and support vector machine for classification.
Specialized in digital signal processing.
- **B. Tech Capstone Project Title:** Automatic Workspace Control and Automation.
- **Key Skills:** Digital signal processing, computer vision, biomedical signal processing, supervised learning, C, C++, Matlab.

Patents Granted [Translated Titles from Korean to English Language] _____

- 2024** Method for intelligent relay node selection for D2D communication, recording medium and device for performing the method. Republic of Korea Intellectual Property Office, 10-2685026.
- 2024** Method for scheduling OFDMA based on quality of service in wireless local area network. Republic of Korea Intellectual Property Office, 10-2648538.
- 2021** Packet collision prevention device and method. Republic of Korea Intellectual Property Office, 10-2276280.
- 2021** Method and apparatus for deep brain stimulation using deep learning and recording medium for performing the method. Republic of Korea Intellectual Property Office, 10-2247761.
- 2020** Apparatus and method for adjusting duty cycle for coexistence of heterogeneous networks and recording medium for performing the method. Republic of Korea Intellectual Property Office, 10-2179791.
- 2020** Method for security mobility adaptable routing for critical IoT and recording medium for performing the method. Republic of Korea Intellectual Property Office, 10-2109771.

Peer-reviewed Publications _____

SCI-INDEXED

- 2025** A. Iqbal, T. Khurshaid, **Y. A. Qadri**, A. Nauman, and S. W. Kim. Priority-Aware Spectrum Management for QoS Optimization in Vehicular IoT in *Sensors*. **SCIE / IF: 3.5**
- 2025** S. Ali, **Y. A. Qadri**, K. Ahmad, Z. Lin, M-F. Leung, S. W. Kim, A.V. Vasilakos, and T. Zhou. Large Language Models in genomics- A perspective on personalized medicine in *Bioengineering*. **SCIE / IF: 3.7**
- 2025** A. Iqbal, A. Nauman, **Y. A. Qadri**, and S. W. Kim. Optimizing Spectral Utilization in Healthcare Internet of Things in *Sensors*. **SCIE / IF: 3.5**
- 2024** **Y. A. Qadri**, K. Ahmad, and S. W. Kim. Artificial General Intelligence for the Detection of Neurodegenerative Disorders in *Sensors*. **SCIE / IF: 3.5**
- 2024** **Y. A. Qadri**, H. Jung, and D. Niyato. Toward the Internet of Medical Things for Real-Time Health Monitoring over Wi-Fi in *IEEE Network*. **SCIE / IF: 6.3**
- 2024** M. A. Jamshed, **Y. A. Qadri**, A. Nauman, H. J. Jung. Electromagnetic Field Exposure-aware AI Framework for Integrated Sensing and Communications-enabled Ambient Backscatter Wireless Networks in *IEEE Internet of Things Journal*. **SCIE / IF: 8.9**
- 2023** A. Nauman, M. A. Jamshed, **Y. A. Qadri**, M. Saad, M. Bilal, V. Shanmuganathan, S. W. Kim. Injecting cognitive intelligence into beyond-5G networks: A MAC layer perspective in *Elsevier Computers and Electrical Engineering*. **SCIE / IF: 4.9**
- 2022** **Y. A. Qadri**, Zulqarnain, A. Nauman, A. Musaddiq, E. Garcia-Villegas, and S.W. Kim. Preparing Wi-Fi 7 for Healthcare Internet-of-Things in *Sensors*. **SCIE / IF: 3.5**
- 2021** Zulqarnain, A. Musaddiq, **Y. A. Qadri**, A. Nauman, M. K. Afzal, and S. W. Kim. RIATA: A Reinforcement Learning-Based Intelligent Routing Update Scheme for Future Generation IoT Networks in *IEEE Access*. **SCIE / IF: 3.6**

- 2021** A. Nauman, M. A. Jamshed, **Y. A. Qadri**, R. Ali, and S. W. Kim. Reliability Optimization in Narrowband Device-to-Device Communication for 5G and Beyond 5G Networks in *IEEE Access*. **SCIE / IF: 3.6**
- 2020** A. Musaddiq, Z. Nain, **Y. A. Qadri**, R. Ali, and S. W. Kim. Reinforcement learning-enabled cross-layer optimization for low-power and lossy networks under heterogeneous traffic patterns in *Sensors*. **SCIE / IF: 3.5**
- 2020** **Y.A. Qadri**, A. Nauman, Y. B. Zikria, A. Vasilakos, and S. W. Kim. The Future of Healthcare Internet of Things: A Survey of Emerging Technologies in *IEEE Communications Surveys & Tutorials*. **SCIE / IF: 46.7**
- 2020** **Y.A. Qadri**, R. Ali, A. Musaddiq, F. Al-Turjman, D. W. Kim, and S. W. Kim. The limitations in the state-of-the-art counter-measures against the security threats in H-IoT in *Springer Cluster Computing*. **SCIE / IF: 4.1**
- 2020** A. Nauman, **Y. A. Qadri**, M. Amjad, Y. B. Zikria, M. K. Afzal, and S. W. Kim. Multimedia Internet of Things: A Comprehensive Survey *IEEE Access*. in **SCIE / IF: 3.6**
- 2019** R. Ali, **Y. A. Qadri**, Y. B. Zikria, T. Umer, B. Kim, and S. W. Kim. Q-learning-enabled channel access in next-generation dense wireless networks for IoT-based eHealth systems in *Springer EURASIP Journal on Wireless Communications and Networking*. **SCIE / IF: 1.9**

Book Chapters

- 2025** **Y. A. Qadri**, A. Nauman, Zulqarnain, R. Ali, and S. W. Kim. **Cobots for Human Surgery**, *Industry 5.0*, DeGruyter.
- 2021** A. Nauman, **Y. A. Qadri**, R. Ali, and S. W. Kim. **Machine learning-enabled Internet of Things for medical informatics**, *Machine Learning, Big Data, and IoT for Medical Informatics*, Elsevier.
- 2020** R. Ali, **Y. A. Qadri**, Y. B. Zikria, F. A. Turjman, B. S. Kim, and S. W. Kim. **A Blockchain Model for Trustworthiness in the Internet of Things (IoT)-Based Smart-Cities**, *Trends in Cloud-based IoT*, Springer.
- 2020** A. Musaddiq, R. Ali, R. Bajracharya, **Y. A. Qadri**, and S. W. Kim. **Trends, Issues, and Challenges in the Domain of IoT-Based Vehicular Cloud Network**, *Unmanned Aerial Vehicles in Smart Cities*, Springer.

Presentations

*presenting author + undergraduate mentee † best paper award

INTERNATIONAL CONFERENCE PRESENTATIONS

- 2024** W. Khalid*, H. Yu, and **Y. A. Qadri**. Active Reconfigurable Intelligent Surface-Aided Terahertz Wireless Communications in *Proceedings of Symposium of the Korean Institute of Communications and Information Sciences, Jeju-do, Republic of Korea*.
- 2022** **Y. A. Qadri***, A. Nauman, Z. Nain, and S. W. Kim. Industrial IoT: Role of IEEE 802.11be WLANs at *95th IEEE Vehicular Technology Conference-Spring, Helsinki, Finland*.
- 2021** Z. Nain*, A. Musaddiq, **Y. A. Qadri**, and S. W. Kim. History-aware Adaptive Route Update Scheme for Low-Power and Lossy Networks at *KICS International Conference on ICT Convergence, Jeju-do, Republic of Korea*.
- 2021** **Y. A. Qadri**, S. H. Park**, and S. W. Kim. Self-optimized channel access for healthcare-Internet of Things at *18th International Workshop on Emerging ICT, Dalian, China*.
- 2019** A. Nauman*, M. A. Jamshed, **Y. A. Qadri**, R. Ali, Y. B. Zikria, and S. W. Kim. An intelligent deterministic D2D communication in narrow-band Internet of Things at *International Wireless Communications and Mobile Computing Conference, Tangier, Morocco*.
- 2018** **Y. A. Qadri***, A. Musaddiq, D. W. Kim, and S. W. Kim. An overview of selective forwarding and wormhole attacks in healthcare IoT at *Asia Pacific Conference on Information Communication Technology, Daegu, Republic of Korea*.
- 2018** A. Musaddiq*, **Y. A. Qadri**, D. W. Kim, and S. W. Kim. An overview of interoperability issues in the vehicular cloud network at *Asia Pacific Conference on Information Communication Technology, Daegu, Republic of Korea*.
- 2016** **Y. A. Qadri*†**, and Vineet Kumar. Early Detection of Epilepsy using Automatic Speech Recognition at *3rd International Conference on Computing Sciences, Jalandhar, India*. Proceedings: Indian Journal of Science and Technology, Vol 9(47), (2016).

INVITED TALKS

Jun. 7, 2023 The Future of Wireless Communication: Enhancing Performance and Efficiency with Innovative Algorithms at Vellore Institute of Technology, India.

Awards

- 2022 **Fully supported research internship between July and August 2022**, i2CAT, Barcelona
- 2021 **Excellent achievements in academic and research support projects**, Ministry of Education, Republic of Korea
- 2018-2020 **Graduate School Scholarship**, Yeungnam University
- 2016 **Best Paper-Signal Processing Track**, 3rd International Conference on Computing Sciences, Jalandhar, India
- 2011-2016 **Merit Scholarship**, L.P. University

Academic Services

Courses

- Artificial Intelligence Applications-Fall, 2024
- Special Topics on Machine Learning-Spring, 2025

Reviewer

- IEEE Transactions on Artificial Intelligence
- IEEE Sensors
- IEEE Access
- IEEE Transactions on Consumer Electronics
- Elsevier Computer Communications
- ACM Computing Surveys

Technical Program Committee (TPC) Member

- IEEE Vehicular Technology Conference-Fall, 2024
- IEEE International Conference on Data Science and Advanced Analytics, 2023

References

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